# DSP – LEC – 8 - Ch-3

# CHAPTER - 3

# CRRELATION AND CONVOLUTION

# CHAPTER – 3

# 3.4 CONVOLUTION DESCRIPTION

# LEC. 8- index

## 3.4 Convolution Description

### 3.4.1 Properties of Convolution

### 3.4.2 Circular convolution

### 3.4.3 Fast linear convolution

### 3.4.4 Computational advantages of fast linear convolution

# 3.4 CONVOLUTION DESCRIPTION

- The term convolution describes, amongst other things, how the input to a system interacts with the system to produce the output.

- Generally the system output will be a delayed and attenuated or amplified version of the input.

- It is particularly useful to consider the output from the system owing to an impulse input.

# Sec. 3.4 Cont. - 1

- This is because any input may be represented as a sequence of impulses of different strengths.

- This figure shows that at sampling instant **"m"** the output owing to the unit impulse applied at sampling instant **0** is h(m).

- The characteristic is known as **the impulse response h(m) of the system.**

# Sec. 3.4 Cont. - 2

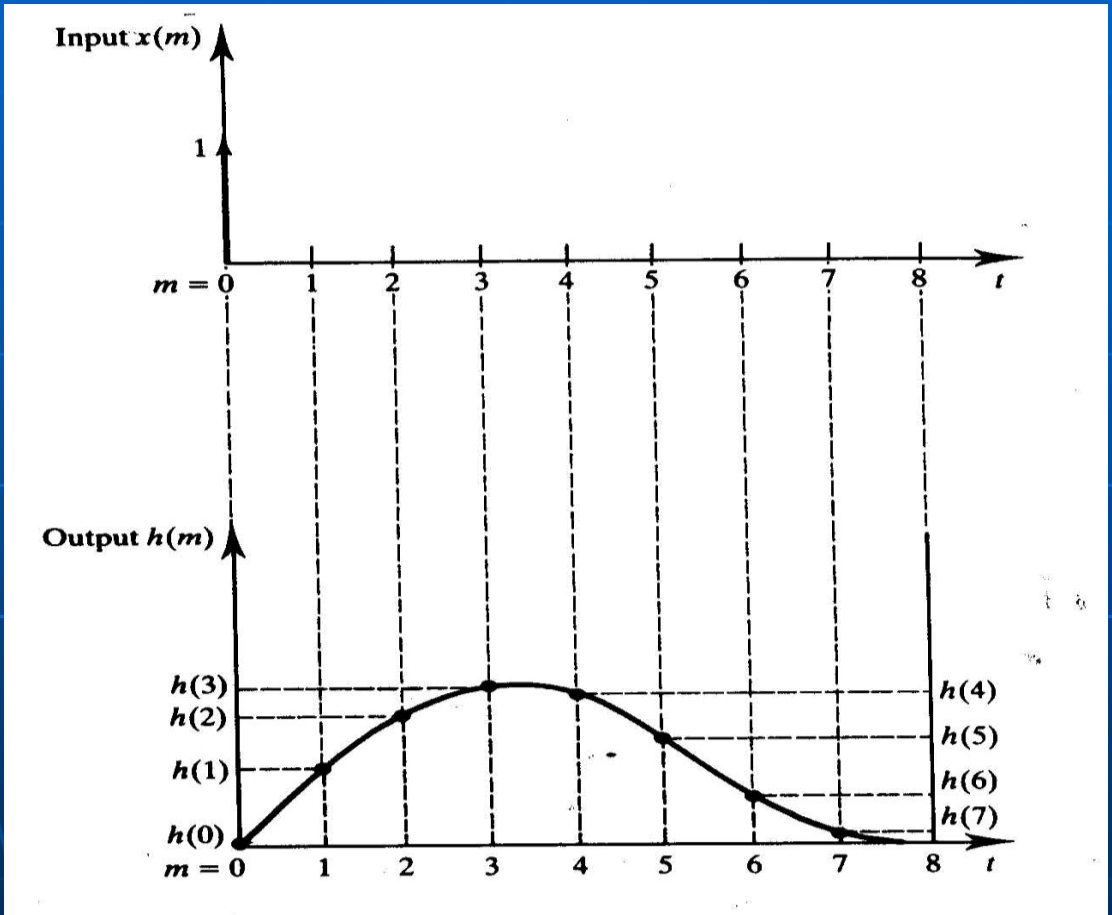



Fig. 3.10 Impulse input and corresponding impulse response of a system

# Sec. 3.4 Cont. - 3

Consider now the application of a sequence of impulses **x(m)** to the system, applied at the sampling instants **m (Fig. 3.11)**.

- Referring to **Fig. 3.11**,
the output at **sampling instant 0** is **y(0)** given by:

$$y(0) = h(0)x(0)$$

# Sec. 3.4 Cont. - 4

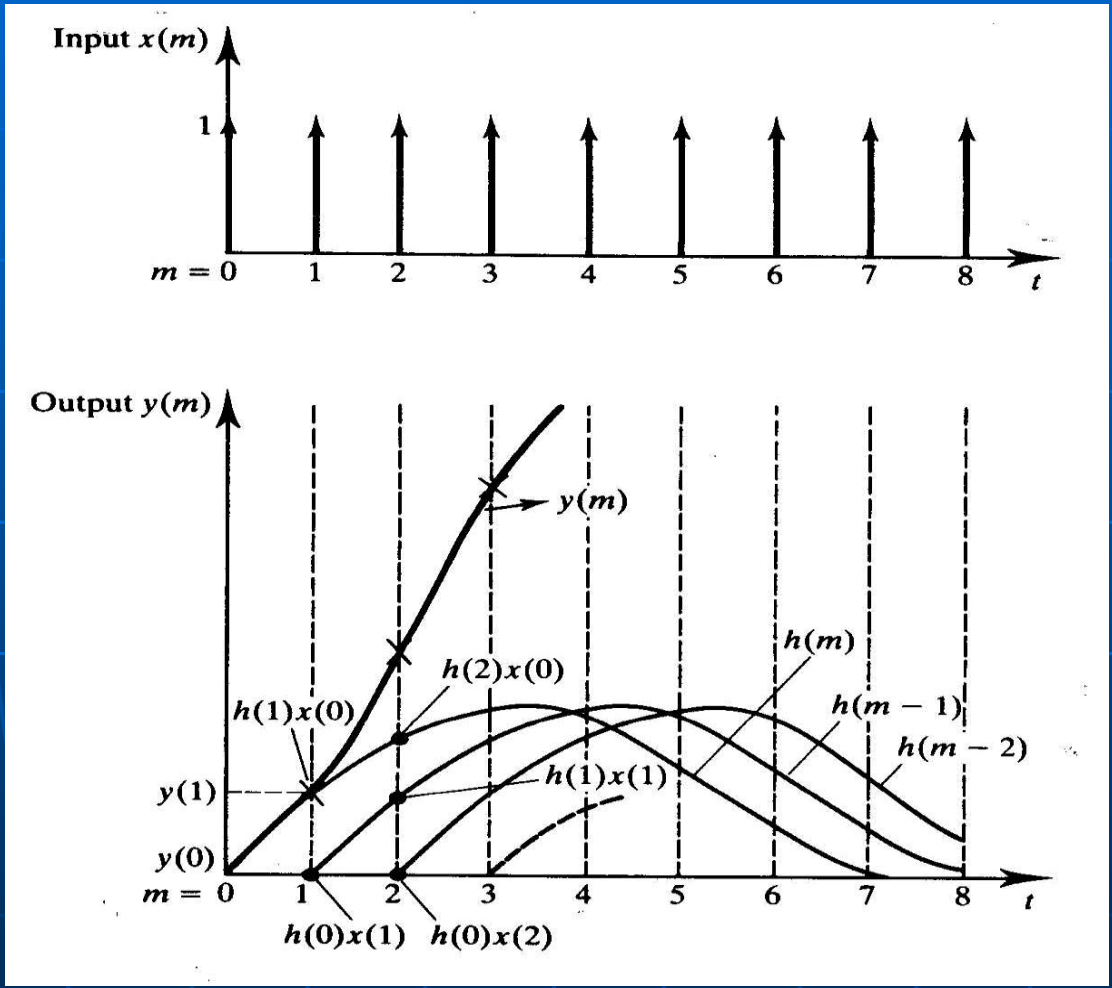


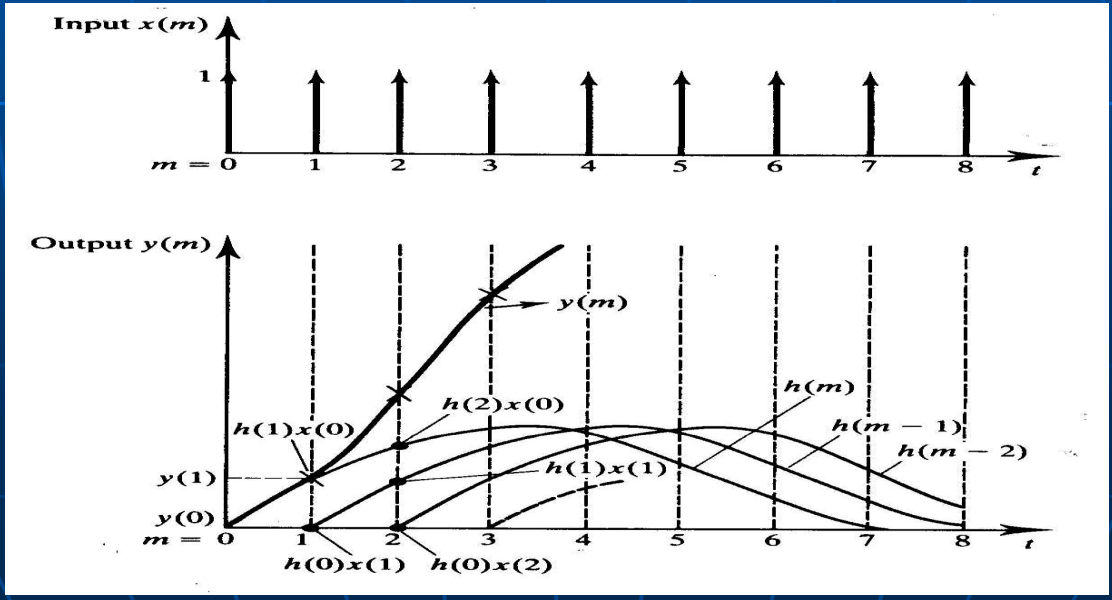

Fig. 3.11 Applied impulse sequence and system response derived from the individual impulse responses.

# Sec. 3.4 Cont. - 5

- At **m = 1**

the output, **y(1),** will be given by h(0)x(1), the effect of the current input x(1), **plus** the delayed effect h(1)x(0) of the input applied at sampling instant **m = 0**. Thus:

$$y(1) = h(1)x(0) + h(0)x(1)$$

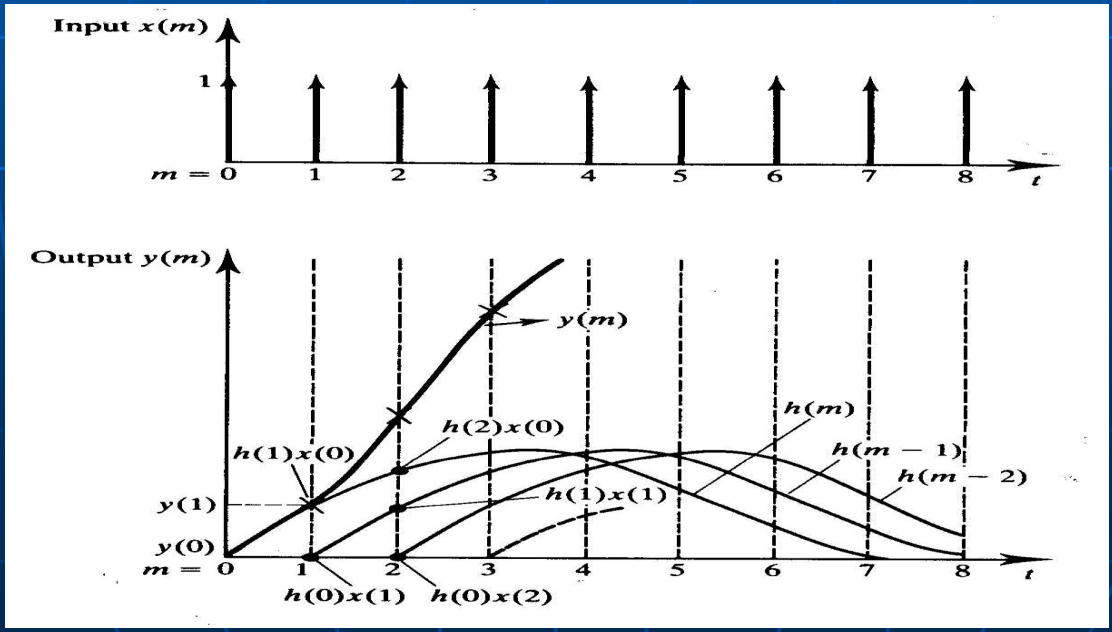


# Sec. 3.4 Cont. - 6

- At **m = 2**

the output, **y(2),** will be given by:

$$y(2) = h(2)x(0) + h(1)x(1) + h(0)x(2)$$



# Sec. 3.4 Cont. - 7

- Similarly, subsequent outputs will be given by:

$$y(3) = h(3)x(0) + h(2)x(1) + h(1)x(2) + h(0)x(3)$$

.

.

$$y(n) = h(n)x(0) + h(n-1)x(1) + h(n-2)\,x(2) + \ldots + h(0)x(n) \quad (3.16)$$

- Alternatively, Eq. **3.16** may be written as:

$$\mathbf{y(n)} = h(0)x(n) + h(1)x(n-1) + h(2)\,x(n-2) + \ldots + h(n)x(0) \quad (3.17)$$

# Sec. 3.4 Cont. - 8

- The output may only be written in this manner as a linear sum of the effects of previous inputs if the system is a **linear** one.

- Equation **3.16** describes the output of a first-order linear system.

$$y(n) = h(n)x(0)+h(n-1)x(1)+h(n-2)\,x(2)+\ldots..+\,h(0)x(n) \qquad (3.16)$$

# Sec. 3.4 Cont. - 9

- The output may be regarded as the product of the corresponding pairs of points in the impulse response function by the time-reversed input sequence.

- **Thus, the convolution sum is equivalent to the cross-correlation of the one sequence by the time-reversed second sequence.**

# Sec. 3.4 Cont. - 10

- Equations **3.16** and **3.17** may be written compactly as:

$$y(n) = \sum_{m=0}^{n} h(n-m)x(m) \qquad (3.18)$$

**or**

$$y(n) = \sum_{m=0}^{n} h(m)x(n-m) \qquad (3.19)$$

- These functions are referred to as the convolution sums of the inputs by the impulse response function.

- **i. e., the output is given by the convolution of the input by the impulse response of the system.**

# Sec. 3.4 Cont. - 11

- Equations **3.18** and **3.19** may be extended to waveforms of infinite duration by writing them as:

$$y(n) = \sum_{m=-\infty}^{\infty} x(m)h(n-m) = x(n) \otimes h(n) \qquad (3.20)$$

or

$$y(n) = \sum_{m=-\infty}^{\infty} h(m)x(n-m) = h(n) \otimes x(n) \qquad (3.21)$$

the generalized form of the convolution sum.

- The symbol $\otimes$ signifies the operation of convolution.

# Sec. 3.4 Cont. - 12

- If the input consists of a **continuous** sequence of impulses the above summations may be replaced by integrals, so, for example, **Eq. 3.20** becomes

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)\, h(t - \lambda)\, d\lambda \qquad (3.33)$$

which is known as **the convolution integral**.

# Sec. 3.4 Cont. - 13

- As shown in previous part, the convolution was carried out in time. This is known as convolution in the time domain.

- **In frequency domain** the output component of a system at frequency, f, is Y(f), given by:

$$Y(f) = H(f)\, X(f) \qquad (3.f\text{-}1)$$

# Sec. 3.4 Cont. - 14

$$Y(f) = H(f)\,X(f) \qquad \textbf{(3.f-1)}$$

Where..

H(f)… is the frequency response function of the system at frequency, f, and

X(f)… is the Fourier transform of the input x(t).

- It can also be consider H(f) is the Fourier transform of h(t).

- The inverse Fourier transform of Eq. **3.f-1** is:

$$F^{-1}[Y(f)] = y(t) = F^{-1}[H(f)X(f)] \qquad \textbf{(3.f-2)}$$

# Sec. 3.4 Cont. - 555

## Example 3.5

# Sec. 3.4 Cont. - 555

**Example 3.5**

Find the convolution y(n) of h(n) and x(n), defined below:

**h(n)** = {4, 3, 2, 1} and **x(n)** = {1, 2, 3, 4}

**Solution**

- The two periodic time sequences **h(m)** = {4, 3, 2, 1} and **x(m)** = {1, 2, 3, 4} will now be convolved.

- Figure **3.12(a)** shows the periodic sequence **h(m)** = (4,3,2,1) and Fig. **3.12(b)** the time-reversed sequence **h(-m)** = (1, 2, 3, 4).

# Example 3.5 - Cont - 1

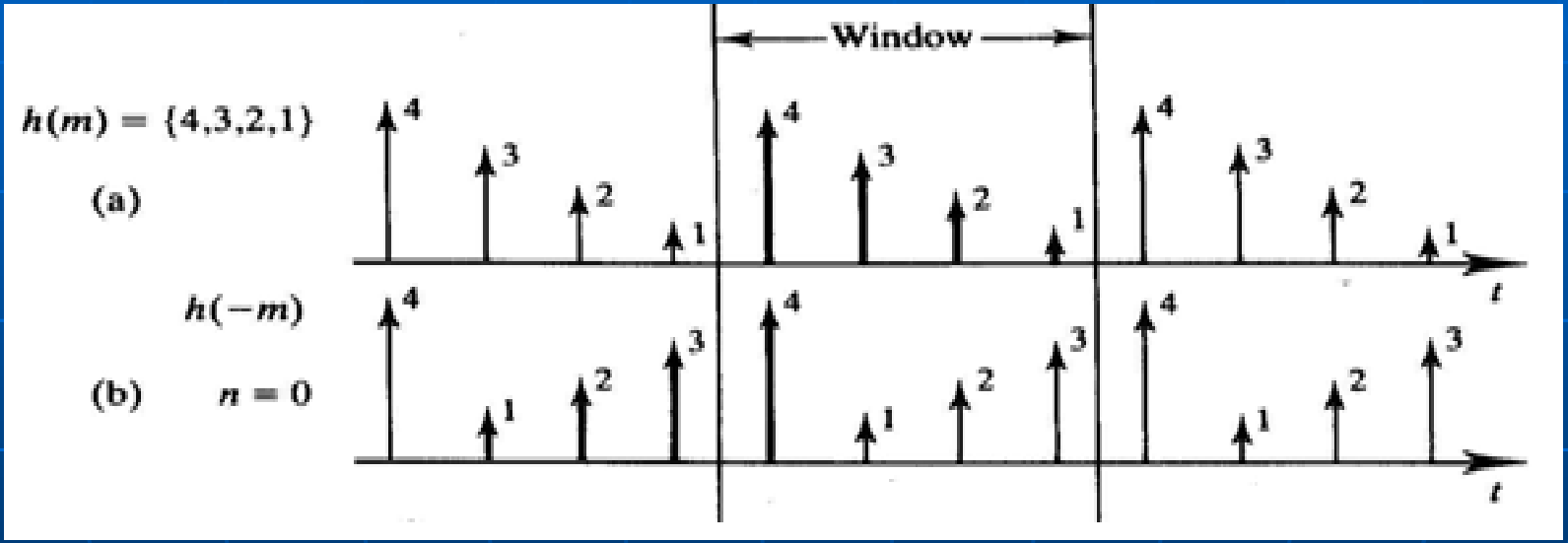



Fig. 3.12 a & b h(m) and h(-m).

## Example 3.5 - Cont - 2

- The convolution sum requires that one sequence has to be multiplied point by point by the time-reversed second sequence, **i.e.,** it corresponds to the cross-correlation of the one sequence by the time reversal of the other.

- The figure also shows a window of width equal to **one period** over which the convolution is evaluated.

- The result obtained will be **periodic** as for the corresponding case of cyclic correlation (**Section 3.2**) and so it is only necessary to evaluate the convolution over the windowed interval.

## Example 3.5 - Cont - 3

Fig. 3.12 The convolution y(m) of h(n) and x(n).

**(a)** The periodic sequence h(m).

**(b)** The time-reversed sequence h(-m).

**(c) to (e)** Versions of h(-m) shifted right by incremental lags.

**(f)** The sequence x(m).

**(g)** The output sequence y(m) = h(n) $\otimes$ x(n).

- Figure **3.12(f)** shows, **x(m)** = (1, 2, 3, 4) for reference.

# Example 3.5 - Cont - 3

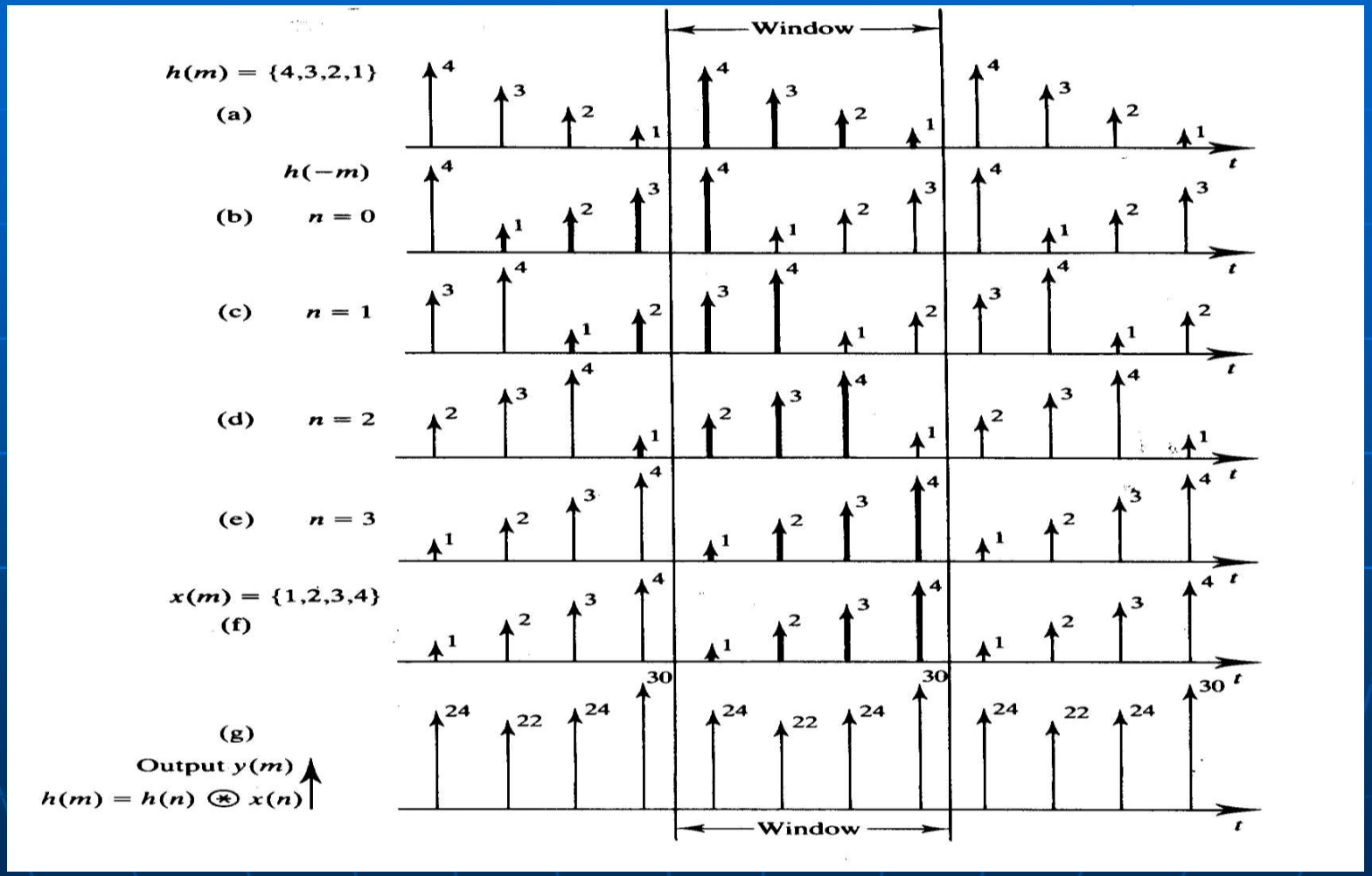



Fig. 3.12 The convolution y(m) of h(n) and x(n).

## Example 3.5 - Cont - 4

Now, when **n = 0**, Eq. 3.31 becomes

$$y(0) = \sum_{m=0}^{n} h(-m)x(m)$$

and is obtained by cross-correlating the windowed data in **Figs. 3.12(b)** and **3.12(f)** :

**y(0)** = 4*1 + 1*2 + 2 *3 + 3 *4 = 24

## Example 3.5 - Cont - 5

When n = 1 Eq. 3-31 becomes

$$y(1) = \sum_{m=0}^{n} h(1-m)x(m)$$

and is obtained by cross-correlating the windowed data in Figures 3.12(c) and 3.12(f), giving

$$y(1) = 3 * 1 + 4 *2 + 1 * 3 + 2 * 4 = 22$$

## Example 3.5 - Cont - 6

Similarly, it is found that

$$y(2) = 2*1 + 3*2 + 4*3 + 1*4 = 24$$

and

$$y(3) = 1*1 + 2*2 + 3*3 + 4*4 = 30$$

So, the convolution y(n) is:

$$y(n) = \{24, 22, 24, 30\}$$

# Example 3.6

**Example 3.6**

Convolve the waveforms x(t) and h(t) of Fig. **3.13(a)** analytically.

**Solution**

Let the convolution integral be

$$y(t) = x(t) \otimes h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau \qquad (3.34)$$

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)\,h(t-\lambda)\,d\lambda \qquad (3.33)$$

- **Equation 3.34** corresponds to **Eq. 3.33** in which the variable $\lambda$ has been replaced by $\tau$ to indicate that time lags are being applied.

## Example 3.6 - Cont - 1

- The convolution integral depends on the variable $\tau$ and so Fig. 3.13(a) has to be replaced by Fig. 3.13(b).

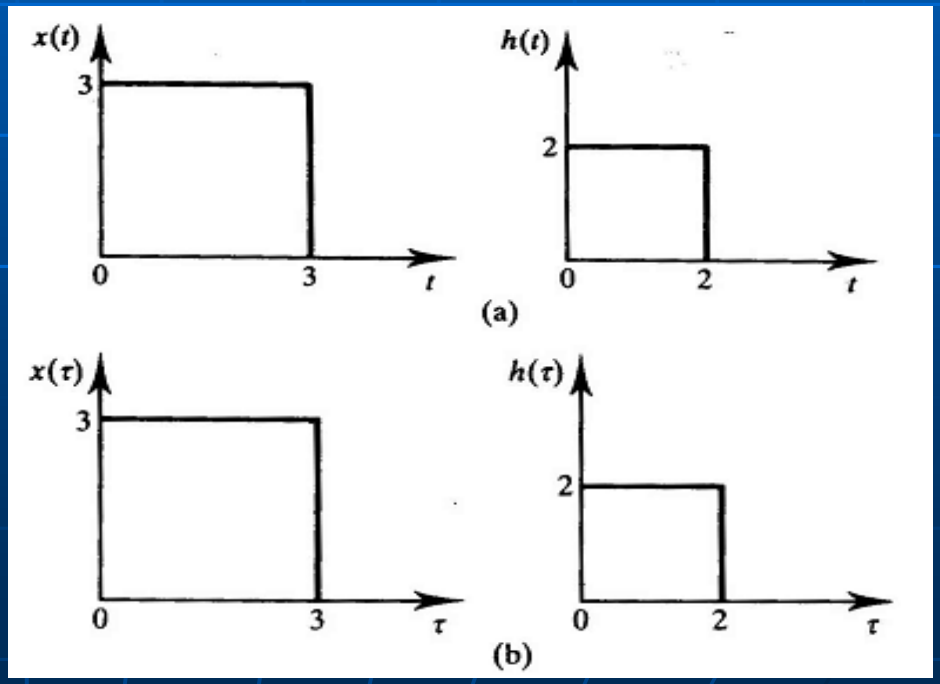
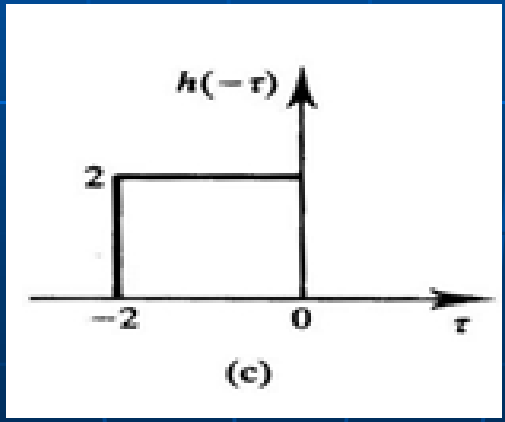



Fig. 3.13
**(a)** The waveforms x(t) and h(t) which are to be convolved analytically.
**(b)** x( $\tau$ ) and h( $\tau$ ) vs $\tau$ .

## Example 3.6 - Cont - 2

- It is now necessary to time reverse h( $\tau$ ) as shown in Fig. **3.13 (c)**.

- h(- $\tau$ ) in next shifted with respect to x( $\tau$ ) in the direction of positive $\tau$.



## Example 3.6 - Cont - 3

- The resulting waveform h(t - $\tau$ ) then overlaps x($\tau$) in **five** separate geometrical stages as shown in **Figs. 3.13(d), 3.13(e & f), 3.13(g), 3.13(h), and 3.13(i)**.

- For each stage there is a corresponding conv. integral.

- Thus x(t) $\otimes$ h (t) exists as five separate regions.

## Example 3.6 - Cont - 4

**Stage 1** ($t < 0$), **Fig. 3.13d**

**Stage 2** ($0 < t \leq 2$), **Fig. 3.13e & f**

**Stage 3** ($2 \leq t \leq 3$), **Fig. 3.13g**

**Stage 4** ($3 \leq t \leq 5$), **Fig. 3.13h**

**Stage 5** ($t > 5$), **Fig. 3.13i**

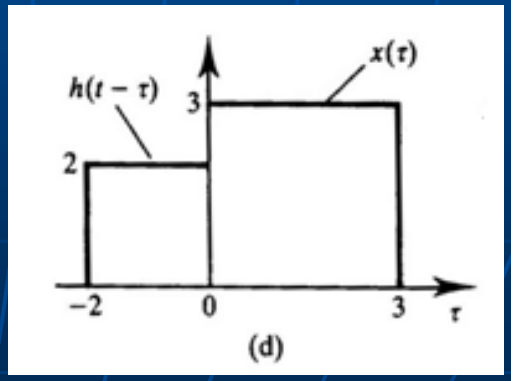
## Example 3.6 - Cont - 5

### Stage 1 (t < 0), Fig. 3.13d

- h(t - $\tau$) does not overlap x($\tau$ ) (Fig. 3.13(d)).

- As the functions do not overlap x($\tau$) h(t - $\tau$) = 0 for all t and there is no contribution to the convolution integral.

**y(t) = 0**



## Example 3.6 - Cont - 6

### Stage 2 ($0 < t \leq 2$), Fig. 3.13e & f

- Partial overlap occurs between $h(t-\tau)$ and $x(\tau)$ (**Fig. 3.13e**).
- Over this range

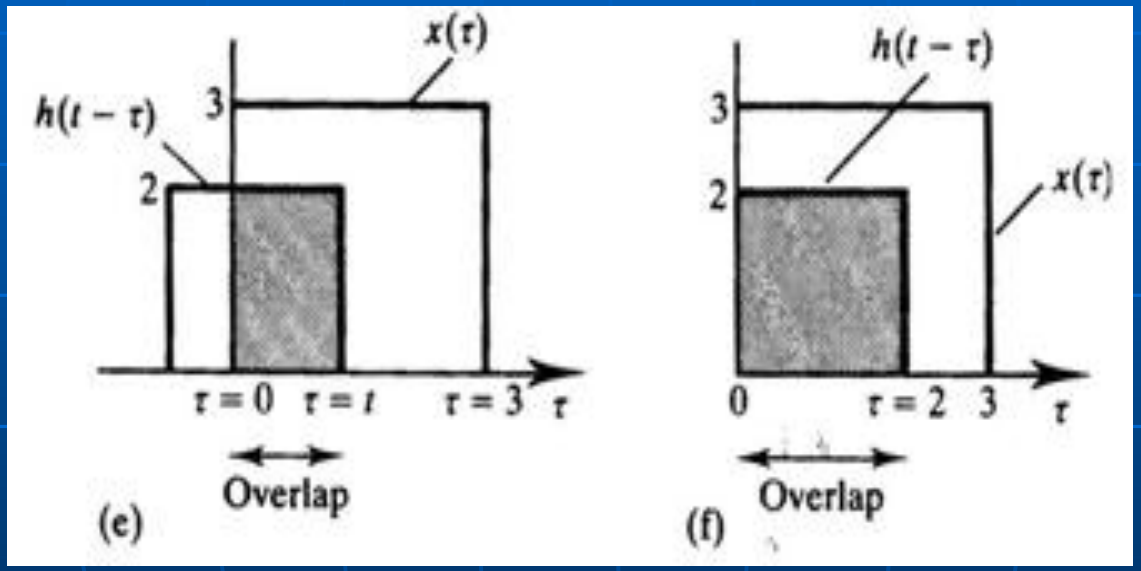
$$y(t) = \int_{\tau=0}^{\tau=t} x(\tau)\, h(t-\tau)\, d\tau = \int_{\tau=0}^{\tau=t} (3)\, x\, (2) d\tau$$

$$y(t) = 6\left[\tau\right]_0^t = 6t, \qquad 0 < t \leq 2 \tag{3.35}$$

This geometrical stage terminates when **t = 2**, as shown in **Fig. 3.13f**).

## Example 3.6 - Cont - 7

### Stage 2 ($0 < t \leq 2$), Fig. 3.13e & f




$$y(t) = 6\left[\tau\right]_0^t = 6t, \qquad 0 < t \leq 2$$

## Example 3.6 - Cont - 8

### Stage 3 ($2 \le t \le 3$), Fig. 3.13g

- There is complete overlap of $h(t - \tau)$ and $x(\tau)$ (**Fig. 3.13g**).

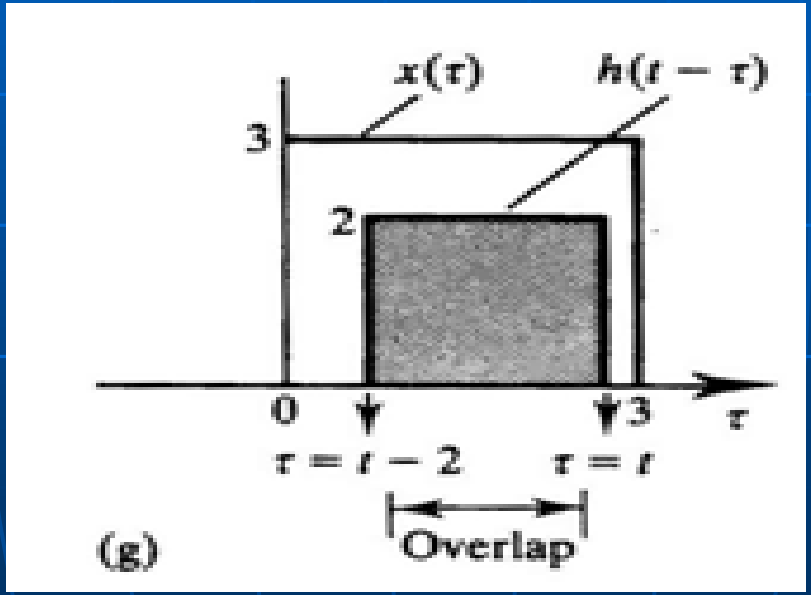
- Over this range of **t**,

$$y(t) = \int_{\tau=t-2}^{t} (3) \times (2) d\tau = 6[\tau]_{t-2}^{t}$$

$$y(t) = 6(t - t + 2) = 12, \qquad 2 \le t \le 3 \tag{3.36}$$

## Example 3.6 - Cont - 9

### Stage 3 ($2 \le t \le 3$), Fig. 3.13g




$$y(t) = 6(t - t + 2) = 12, \qquad 2 \le t \le 3$$

## Example 3.6 - Cont - 10

### Stage 4 (3 ≤ t ≤ 5), Fig. 3.13h

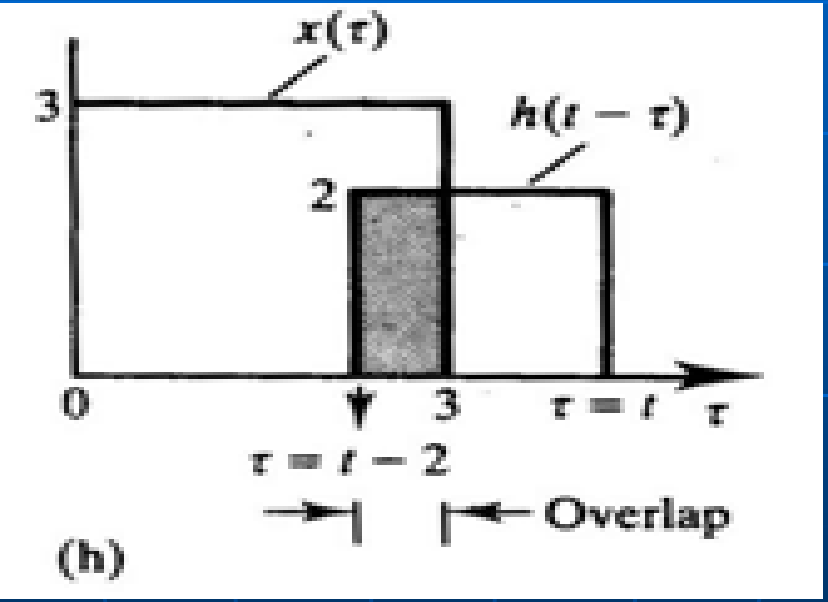
- This case of overlap region is shown in **Fig. 3.13h:**

$$y(t) = \int_{\tau=t-2}^{\tau=3} (3) \times (2)\, d\tau = 6\left[\tau\right]_{t-2}^{3}$$

$$y(t) = 6\,(5 - t) = 30 - 6t, \qquad 3 \le t \le 5 \tag{3.37}$$

## Example 3.6 - Cont - 11

## Stage 4 (3 ≤ t ≤ 5), Fig. 3.13h




$$y(t) = 6\,(5 - t) = 30 - 6t, \qquad 3 \le t \le 5$$

## Example 3.6 - Cont - 12

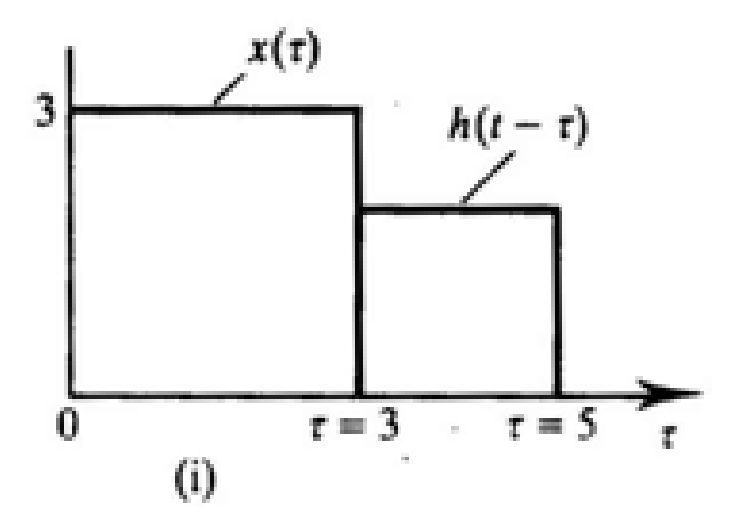
### Stage 5 (t > 5), Fig. 3.13i

- As seen in **Fig. 3.13i,** this is a second region of no overlap, and so there is no contribution to the convolution integral.

$$y(t) = 0$$

## Example 3.6 - Cont - 13

### Stage 5 (t > 5), Fig. 3.13i




$y(t) = 0$

# Example 3.6 - Cont - 14

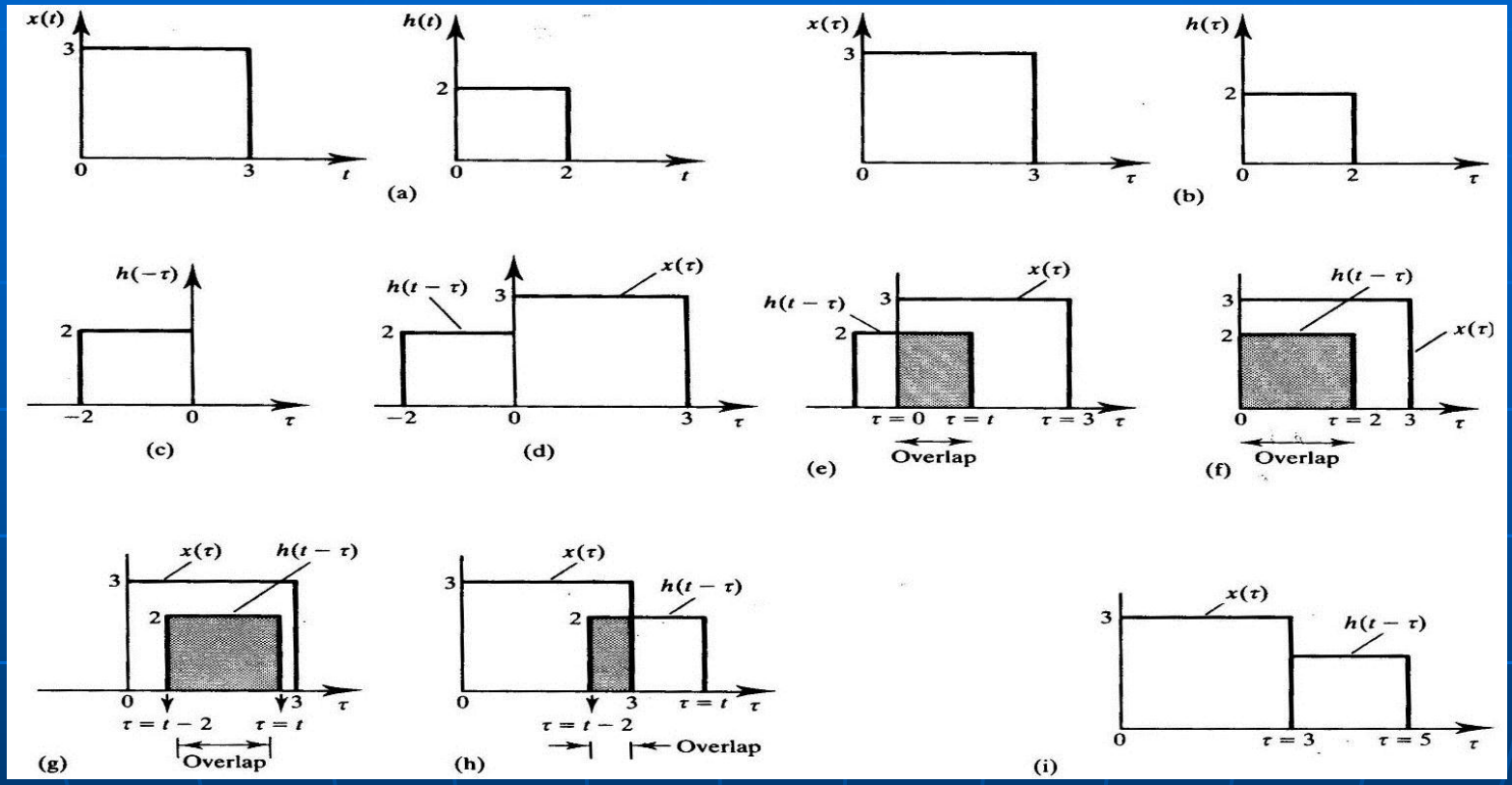



Fig. 3.13 (a) The waveforms x(t) and h(t) which are to be convolved analytically. (b) x(τ) and h(τ) vs τ. (c) h(-τ) vs τ. (d) h(t - τ) and x(τ) vs τ. t < 0, h(t - τ) does not overlap x(τ). (e) h(t - τ) and x(τ) vs τ. 0 < t ≤ 2. First partial overlap occurs between h(t - τ) and x(τ). (f) h(t - τ) and x(τ) vs (τ). t = 2. End of first partial overlap. (g) h(t - τ) and x(τ) vs(τ). 2 ≤ t ≤ 3. Complete overlap between h(t - τ) and x(τ) occurs. (h) h(t - τ) and x(τ) vs (τ). 3 ≤ t ≤ 5. Second partial overlap occurs between h(t - τ) and x(τ). (i) h(t - τ) and x(τ) vs τ. t > 5. h(t- τ) does not overlap x(τ).

## Example 3.6 - Cont - 15

- The convolution integral having a different expression for each of the five regions summarized as shown below:

| | | |
|---|---|---|
| **Stages 1** | $t < 0$ | $y(t) = 0$ |
| **Stages 2** | $0 < t \leq 2$ | $y(t) = 6t$ |
| **Stages 3** | $2 \leq t \leq 3$ | $y(t) = 12$ |
| **Stages 4** | $3 \leq t \leq 5$ | $y(t) = 30 – 6t$ |
| **Stage 5** | $t > 5$ | $y(t) = 0$ |

## Example 3.6 - Cont - 16

**Stages 2 to 4** thus each make a contribution to the convolution integral,

| | | |
|---|---|---|
| **Stages 2** | $0 < t \leq 2$ | $y(t) = 6t$ |
| **Stages 3** | $2 \leq t \leq 3$ | $y(t) = 12$ |
| **Stages 4** | $3 \leq t \leq 5$ | $y(t) = 30 - 6t$ |

From these expressions y(t) may be plotted against t as shown in **Fig. 3.14**.

## Example 3.6 - Cont - 17

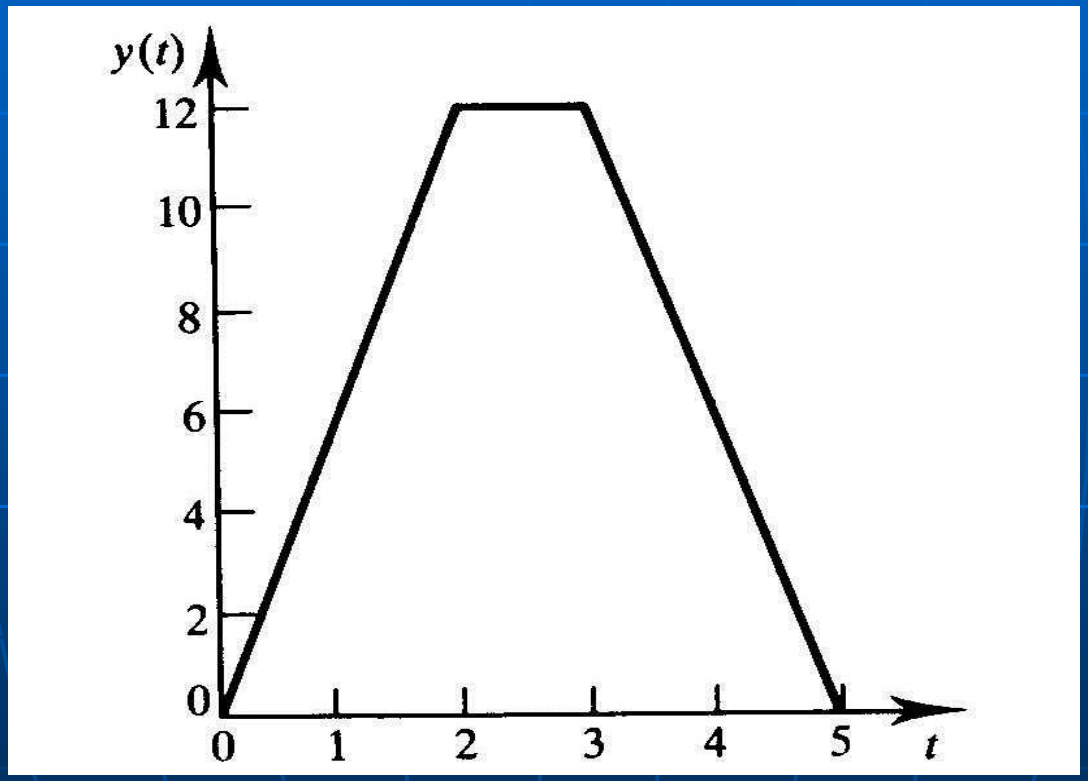



Fig. 3.14 The convolution y(t) = x(t) ⊗ h(t) V.S t.

## 3.4.1. Convolution Properties

### 1- Commutative law:

$$x_1(t) \otimes x_2(t) = x_2(t) \otimes x_1(t) \tag{3.44}$$

**Note that this is identical to:**

$$\int_{-\infty}^{\infty} x_1(\tau) x_2(t-\tau)\, d\tau = \int_{-\infty}^{\infty} x_2(\tau) x_1(t-\tau)\, d\tau$$

# Sec. 3.4 Cont. - 16

**2- Distributive law:**

$$x_1(t) \otimes [x_2(t) + x_3(t)] = x_1(t) \otimes x_2(t) + x_1(t) \otimes x_3(t)$$

**3- Associative law:**

$$x_1(t) \otimes [x_2(t) \otimes x_3(t)] = [x_1(t) \otimes x_2(t)] \otimes x_3(t) \qquad (3.45)$$

## 3.4.2. Circular convolution

- As illustrated previously, the result of correlating two unequal length periodic sequences was a cyclic sequence of period equal to that of the **shorter sequence**, which was, therefore, an incorrect result.

- Because convolution is equivalent to the cross-correlation of one sequence by the reverse of a second sequence, the same will be true of convolution.

- Therefore, as with correlation, in convolution it is necessary that the two sequences be of the same length.

# Sec. 3.4 Cont. - 18

- Thus if the sequence lengths are $N_1$ and $N_2$, then $N_2 - 1$ augmenting zeros must be added to the sequence of length $N_1$, and $N_1 - 1$ augmenting zeros must be add to the sequence of length $N_2$.

- Both sequences will now be of identical length $N_1 + N_2 - 1$, and the correct **linear convolution** will be obtained.

## 3.4.3. Fast linear convolution

- In Section 3.3 it was shown that correlation computations could be **speeded up** using the correlation theorem.

- A similar theorem, the convolution theorem, exists in the case of convolution.

- Thus, in discrete terminology and for the time domain:

$$x_1(l) \otimes x_2(r) = F_D^{-1}[X_1(k)X_2(k)] \qquad (3.46)$$

# Sec. 3.4 Cont. - 20

**Fast linear convolution – cont -1**

$$x_1(\ell) \otimes x_2(r) = F_D^{-1}[X_1(k)X_2(k)] \qquad (3.46)$$

Equation 3.46 is the convolution theorem,

**Where**:

$F_D^{-1}$ ..... denotes the inverse discrete Fourier transform,

$X_1(k)$... is the discrete Fourier transform of $x_1(\ell)$, and

$X_2(k)$... is the discrete Fourier transform of $x_2(r)$.

As in **Section 3.3**, $x_1(\ell)$ and $x_2(r)$ are periodic sequences of length, N.

## 3.4.4 Computational advantages of fast linear convolution

- The method of fast linear convolution only offers the advantage of greater computational speed over the direct approach if the number of values to be convolved is sufficiently large.

- The **number of multiplications** required to perform the convolution by the direct and fast method is compared here as a measure of their relative computational efficiencies.

# Sec. 3.4 Cont. - 22

- The necessary computations for the direct method were given in **Eq. 1**.

$$y(n) = \sum_{m=-\infty}^{\infty} x(m)h(n-m) = x(n) \otimes h(n) \qquad (1)$$

- In this equation, to obtain the linear convolution of the two N-point sequences h(n – m) and x(m) it is necessary to multiply each value of h(n – m) by each value of x(m), making **N*N = $N^2$** multiplications in all.

# Sec. 3.4 Cont. - 23

- The convolution computations could be speeded up using the convolution theorem, defined as:

$$x_1(l) \otimes x_2(r) = F_D^{-1}[X_1(k)X_2(k)] \qquad (2)$$

**Where**:

$F_D^{-1}$ ..... denotes the inverse discrete Fourier transform,

$X_1(k)$... is the discrete Fourier transform of $x_1(\ell)$, and

$X_2(k)$... is the discrete Fourier transform of $x_2(r)$.

# Sec. 3.4 Cont. - 24

- Now consider linear convolution of the same two **N-point** sequences by the fast method according to **Eq. 2**.

- The addition of the necessary augmenting zeros means each sequence is of length **2N-1** points.

- Assume $2N - 1 \approx 2N$ for example $N \geq 8$, and in order to use 2 FFT, **N** is given by an integer power of 2, i.e., $N = 2^d$ where **d** is an integer.

# Sec. 3.4 Cont. - 25

-We know that the number of complex multiplications for

an N-point FFT is **$(N/2)\log_2 N$**,

So for **2N-point FFT (($(2N/2)\log_2 2N$** or

**$N\log_2 2N$)** complex multiplications.

# Sec. 3.4 Cont. - 26

- **Equation-2,** requires **two** **DFTs ( or FFT)** and **one** **IDFT (or IFFT)** to be computed.

$$x_1(l) \otimes x_2(r) = F_D^{-1}[X_1(k)X_2(k)] \qquad (2)$$

- So, it is required to compute **three** **2N – point** **FFTs** involving **$3N \log_2 2N$** complex multiplications.

# Sec. 3.4 Cont. - 27

- Further, for each of the **2N** values of **Eq. 2**, it is necessary to evaluate the complex multiplications **[**$X_1(k)\ X_2(k)$**]**, **thus** increasing the number of complex multiplications necessary to:

$$\mathbf{3N \log_2 2N + 2N.}$$

# Sec. 3.4 Cont. - 28

- Now, each complex multiplication of the form

**(A + jB) (C +jD)**

requires **four** **real** multiplications:

AC, AD, BC and BD.

## Sec. 3.4 Cont. - 29

- So, the number of complex multiplications required in **Eq. 2**,

$$3N \log_2 2N + 2N$$

**becomes:**

$$12N \log_2 2N + 8N$$

real multiplications.

# Sec. 3.4 Cont. - 30

**Table 3.1** Numbers of real multiplications necessary for the convolution of **2N - point** sequences

| N | *Direct method* | Fast convolution | Number ratio, fast : direct |
|---|---|---|---|
| 8 | 64 | 448 | 7 |
| 16 | 256 | 1088 | 4.25 |
| 32 | 1024 | 2560 | 2.5 |
| 64 | 4096 | 5888 | 1.4375 |
| 128 | 16384 | 13312 | 0.8125 |
| 256 | 65536 | 29696 | 0.4531 |
| 512 | 262144 | 65536 | 0.250 |
| 1024 | 1048576 | 143360 | 0.1367 |
| 2048 | 4194304 | 311296 | 0.0742 |